

Frankie Gillis

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Education

University of St Andrews | MMath (Hons) Mathematics Sept 2024 – May 2028

- Honours modules: Linear Mathematics 2 (19.8), Real Analysis (20), Differential Equations (20), Mathematical Statistics (20).
- Sub-honours modules: Linear Mathematics (20), Analysis (20), Multivariate Calculus (19), Combinatorics and Probability (20), Abstract Algebra (20), Vector Calculus (20), Mathematical Modelling (20), Statistical Inference (19).

Research Experience

Research Intern Summer 2025

- Summer research internship under the supervision of Dr Thomas Coleman in the School of Mathematics and Statistics researching partial graph automorphisms and their application to studying pseudo-similar vertices in graphs.
- Received £3,000 funding from the Laidlaw Foundation. Presented my research poster at the Laidlaw Scholars Conference, where it was rated 'highly commended' by a commissioning editor at Taylor and Francis.

Relevant Awards

Tullis Medal and Prize 2025

- 'Awarded to the best student(s) in 2000-level Pure Mathematics.'

Writings

"Understanding Pseudo-Similarity in Graphs: a Path to Proving the Reconstruction Conjecture", Research Essay, Published on the Laidlaw Scholars Network.

"Understanding Pseudo-similarity in Graphs: a Path to Proving the Reconstruction Conjecture", Poster, Presented at the Laidlaw Scholars Conference, Durham University, October 2025.

Additional Experience

Software Contributor Feb 2025 – Present

- Contributed to the development of the Digraphs package for the GAP computer algebra system, for computing with directed graphs. Part of the vertically-integrated project in computational mathematics supervised by Prof James Mitchell.
- Designed and implemented an algorithm to determine if a digraph is 2-edge transitive. Reduced the time complexity of enumerating the 2-edges from $\mathcal{O}(n^3)$ to $\mathcal{O}(n^2 + m)$ for a digraph with n vertices and m edges. Utilised the Orbit-Stabiliser theorem in computing the final step.
- Implemented the method `DigraphMinimumCutSet` to find the minimal cut of a network using the max-flow min-cut theorem from combinatorial optimisation, utilising the existing method `DigraphMaximumFlow`.

Outreach

Assistant Instructor | The Timothy Smith Network | Boston, USA Summer 2026

- Assistant instructor for the Timothy Smith Network's Boston Design Academy, teaching HTML, CSS and web design principles to underrepresented groups in the Boston area. Received £3,000 funding from the Laidlaw Foundation, and a further £2,000 for travel expenses.

Skills and Interests

Software: Python, $\LaTeX$, GAP.